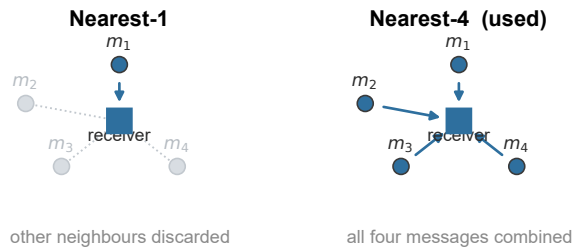


(a)

Message aggregation

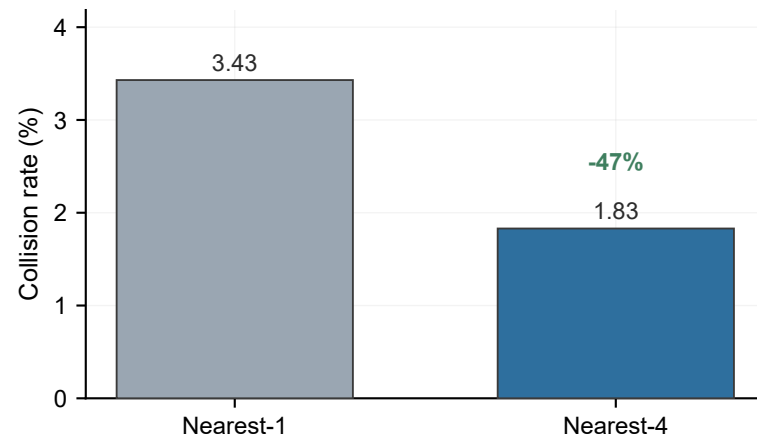


$$[\sin \phi_{ij}, \cos \phi_{ij}, d_{ij}/420] \parallel \mathbf{m}_j \rightarrow f_{\text{loc}} \rightarrow \mathbf{e}_{ij}$$

$$\hat{\mathbf{m}}_i = \frac{1}{|P_i|} \sum_{j \in P_i} \mathbf{e}_{ij} \quad (\text{masked mean})$$

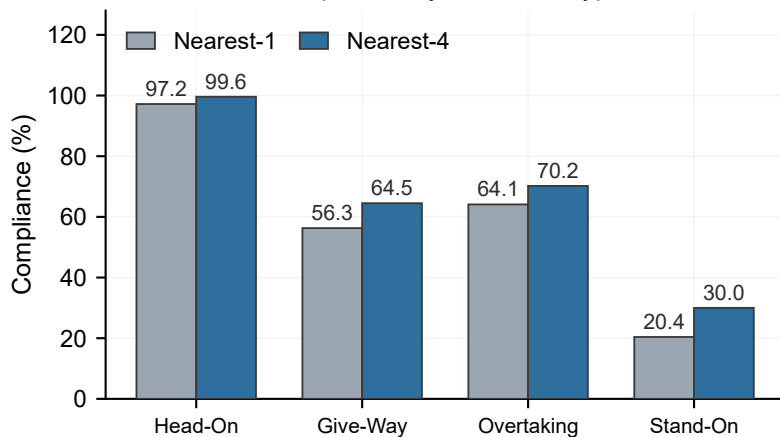
(b)

Collision rate



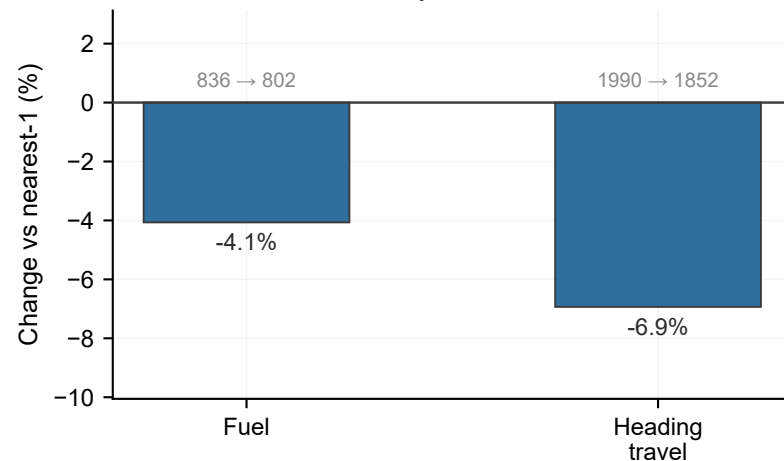
(c)

Compliance by encounter type



(d)

Efficiency vs nearest-1



Arrival rate is 55.0% → 55.6% and is reported in the text. Overall compliance 45.6% → 52.3%.
A pairwise communication bottleneck fails when resolving one encounter changes the geometry of another.